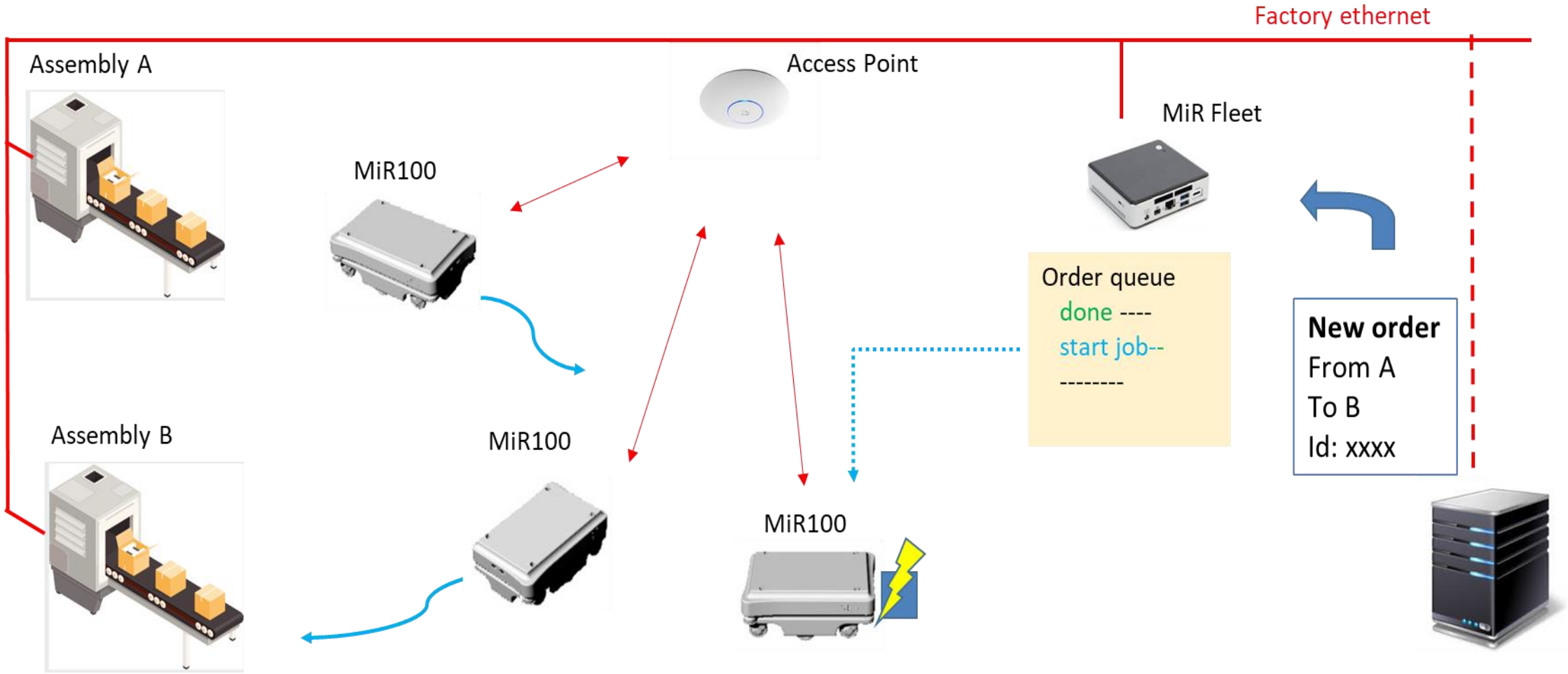
MiR API Training

11 January 2022

ReST API application example



Structure of a request - MiR

METHOD (HTTP method) -> MiR uses the following four main HTTP methods:

GET, to retrieve data
PUT, to update existing data
POST, to send new data
DELETE, to remove existing data

IP ADDRESS

IP address of MiR robot or MiR Fleet

API VERSION

the actual MiR API version is v2.0.0

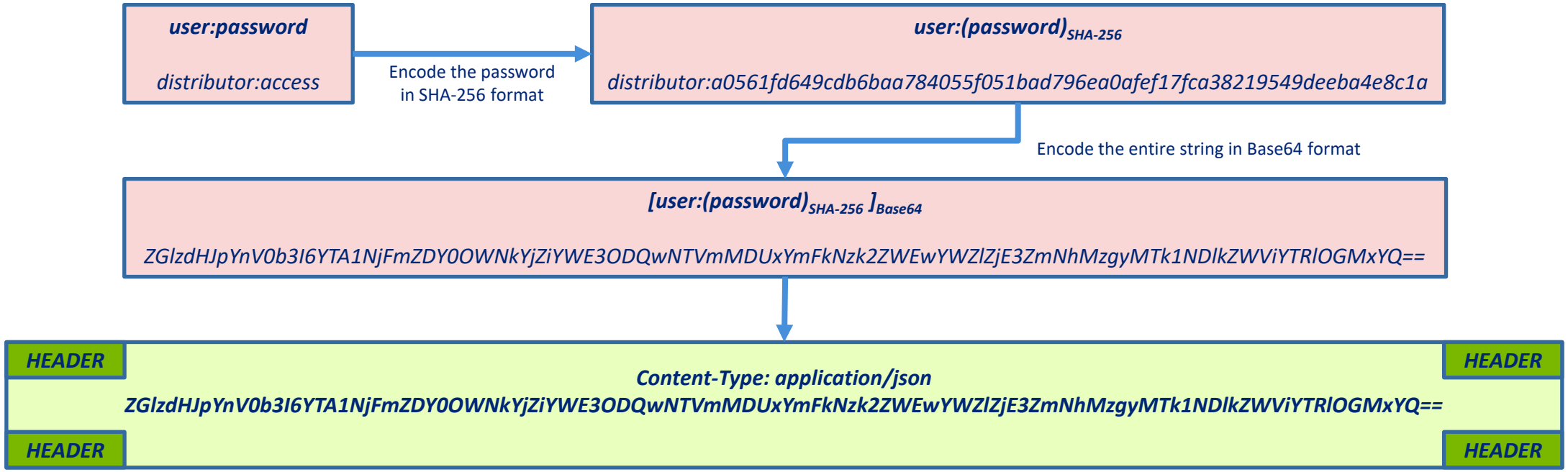
ENDPOINT

MiR provides all the endpoint in the documentation. A few examples: maps, status, missions, etc.

Structure of a request/Security - MiR

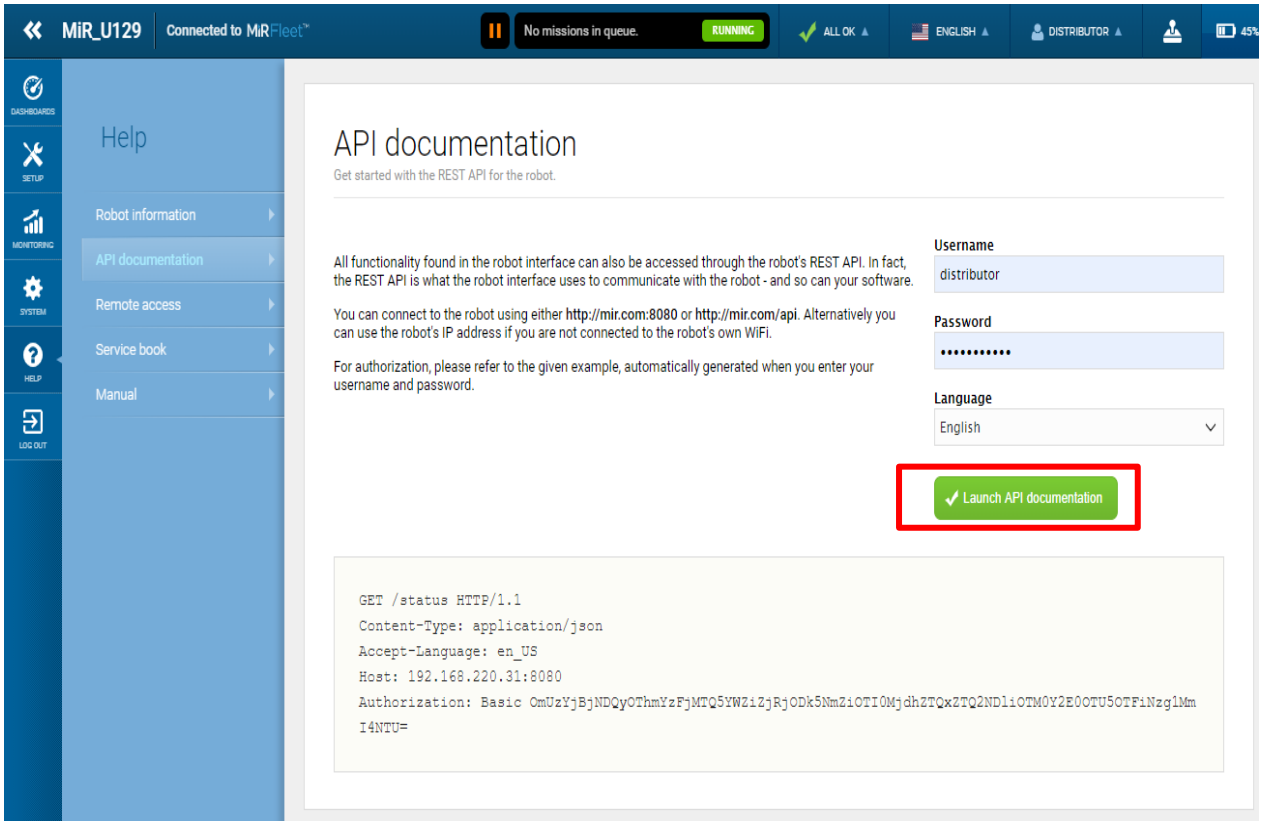
- The REST API communication with MiR robot uses encryption,
  based on *the username and the password of one of the users defined in the robot UI*:
  this results in a *header* for the API request which must be declared in a specific way.
  In particular, the header of the API request must contain the encryption of the line ***user:password***

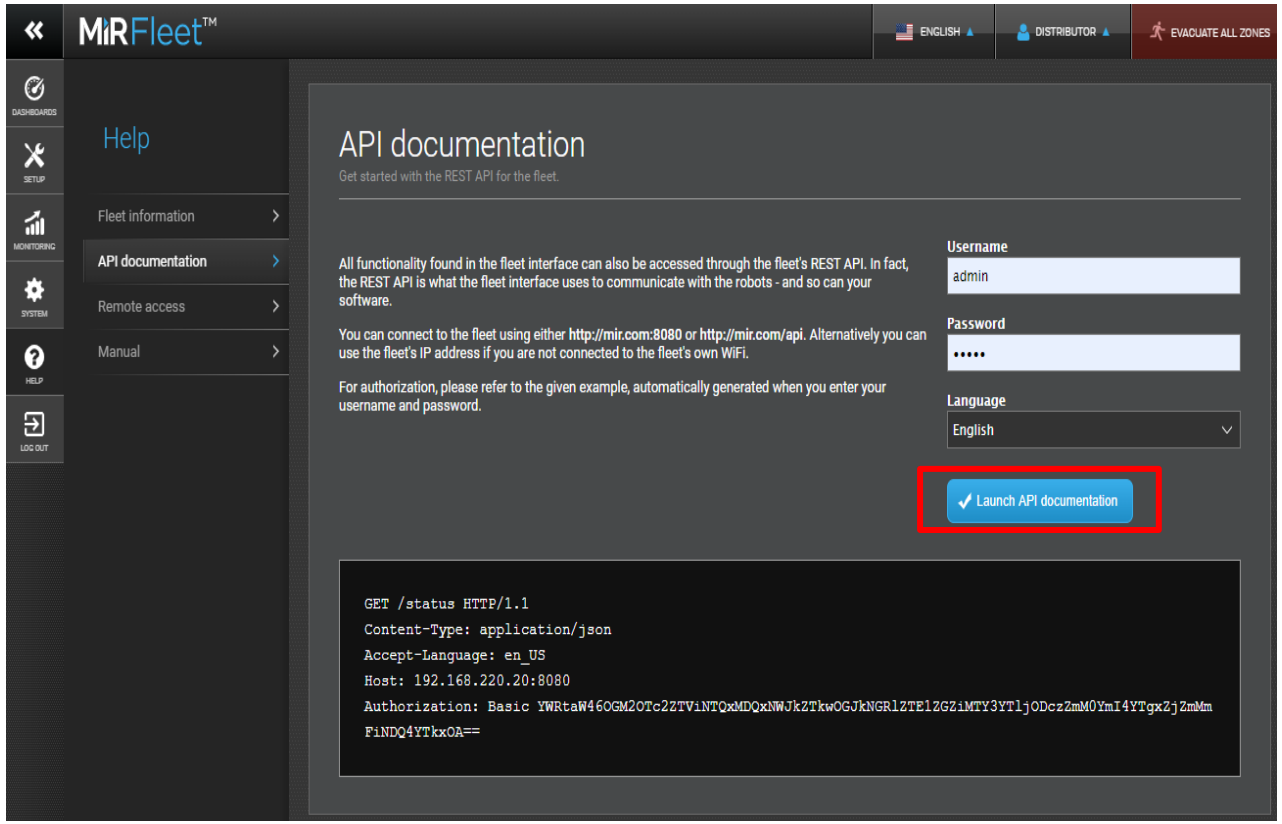
- Considering the user *distributor*, which in this example has password *"access"*, the process is:



Getting Start

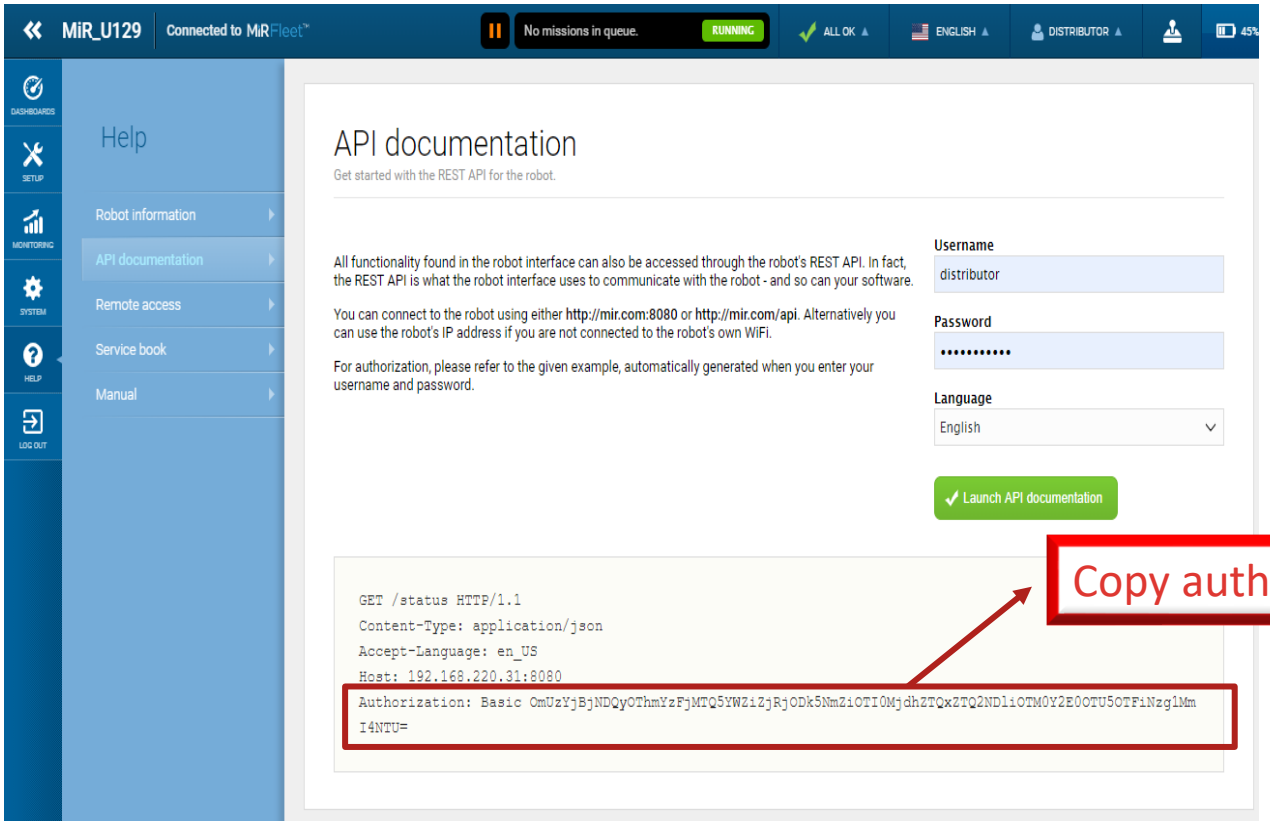
Go to **Help > API documentation**



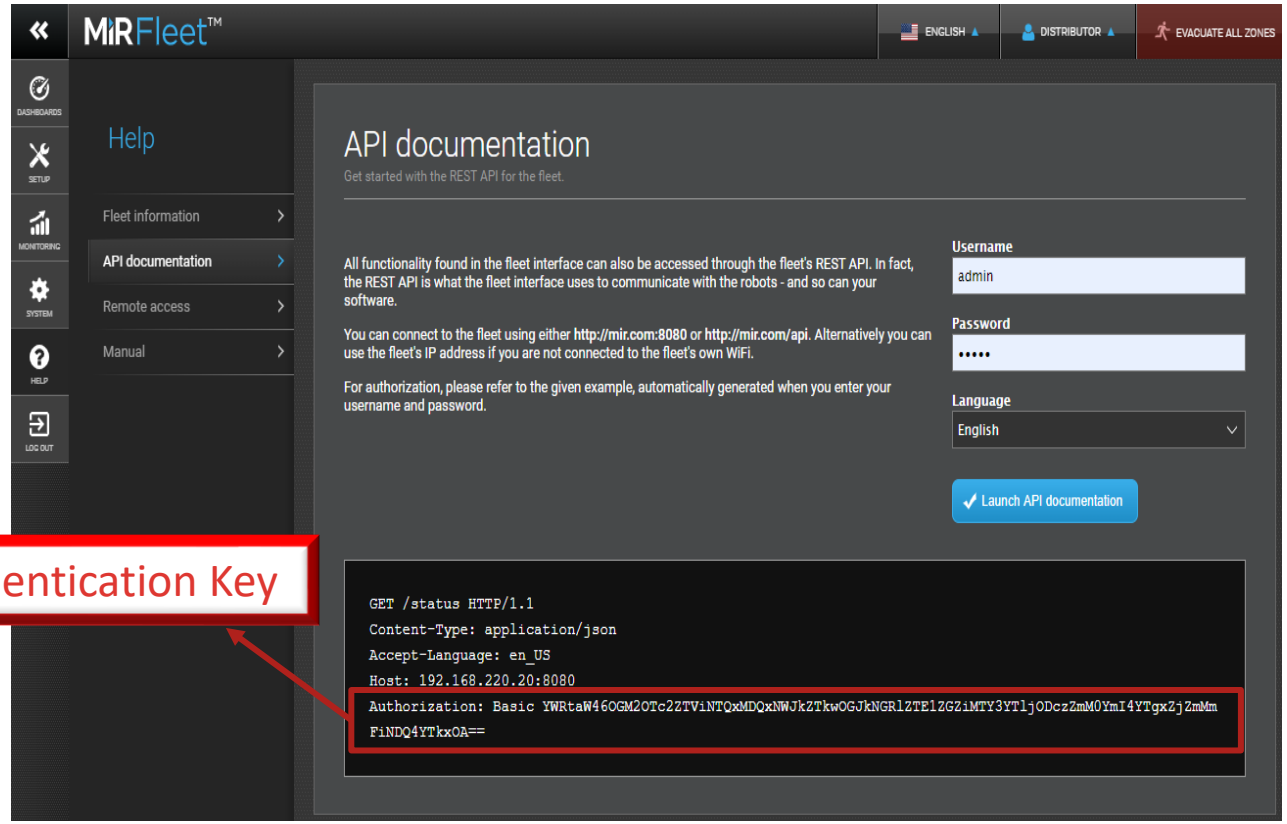


Getting Start

Go to **Help > API documentation**

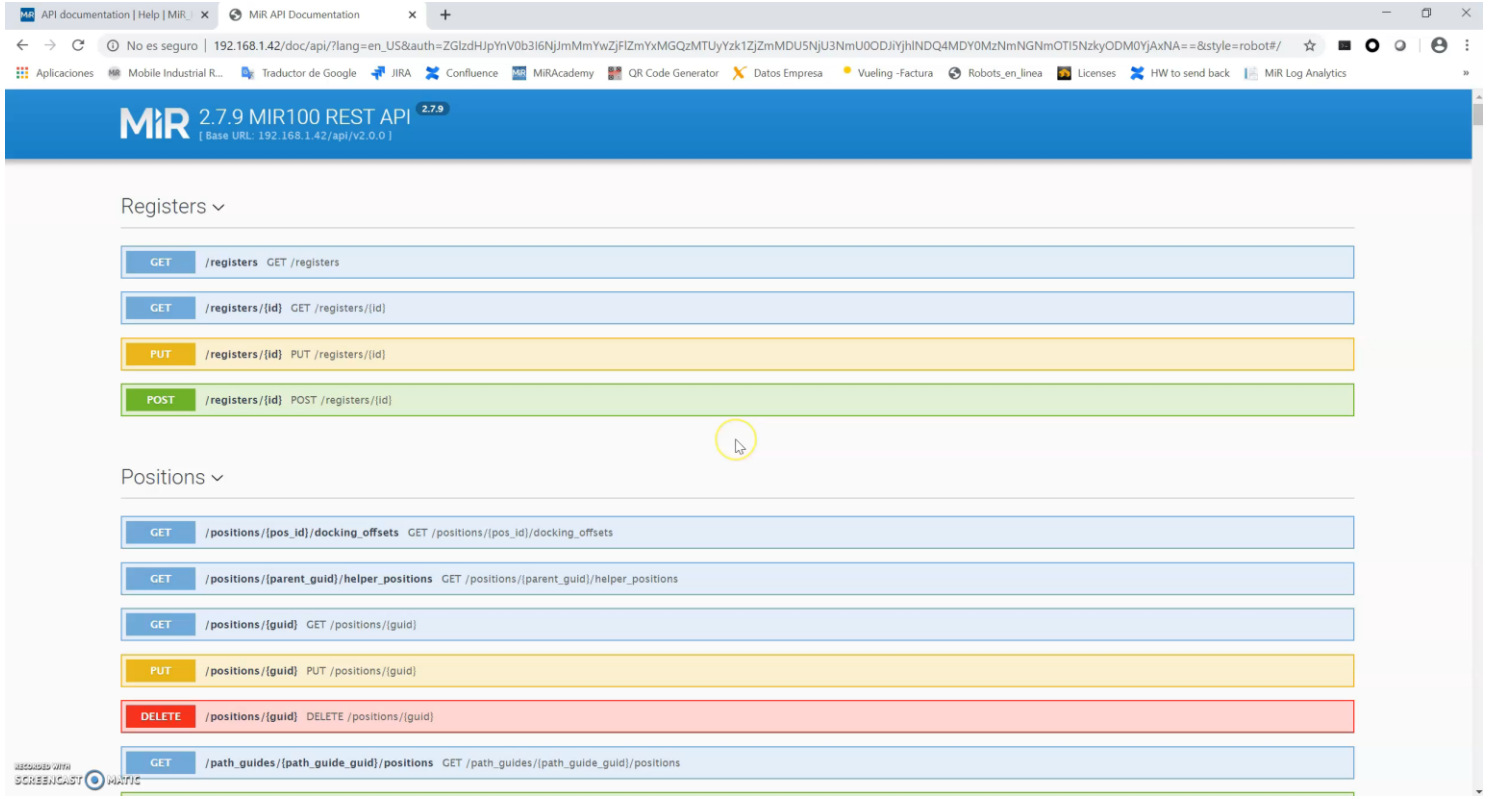


Copy authentication Key



API Documentation

- *API documentation* not only lists all the available REST APIs but gives also the possibility to execute them.
  In the following example, the STATUS of the robot is gathered:



API Document

POST /mission_queue POST /mission_queue

[IP]/api/v2.0.0/

Add a new mission to the mission queue. The mission will always go to the end of the queue

Parameters

Name	Description
Authorization * required string *(header)*	Authorization header
Accept-Language * required string *(header)*	Language header
Mission_queues * required *(body)*	The details of the mission_queues

Example Value | Model

```
{
  "mission_id": "string",
  "message": "string",
  "parameters": [
    {}
  ],
  "priority": 0
}
```

Body

Parameter content type

application/json

Example Value | Model

```
PostMission_queues {
    mission_id*     string
    message         string
                    Max length: 200
    parameters      [ {
                    }]
    priority        integer($float)
}
```

Check required field

Getting Started with Postman

MiR

Getting Start with Postman

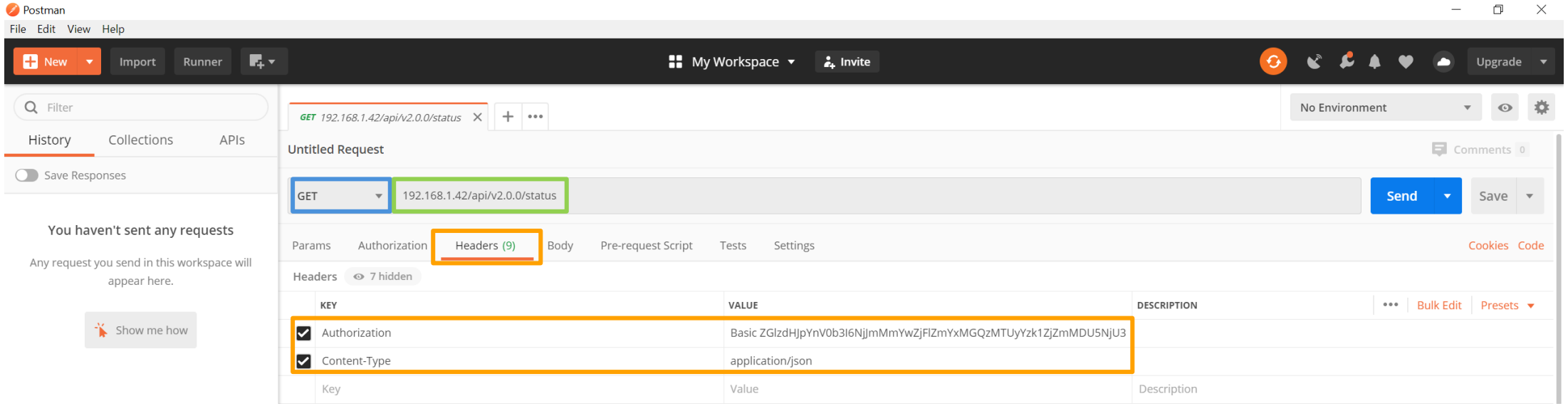
POSTMAN

Download Link: https://www.postman.com/downloads/

- *Postman* is one of the free development tools that can be used to test REST API communication with MiR

Getting Start with Postman

- The Postman window shows:



HEADER

HTTP METHOD

IP ADDRESS / API VERSION / ENDPOINT

Getting Start with Postman

PUT | http://192.168.220.30/api/v2.0.0/status

Params | Authorization | Headers (2) | Body | Pre-request Script | Tests

	KEY	VALUE
☑	Authorization	Basic YWRtaW46OGM2OTc2ZTViNTQxMDQxNWJkZTkwOGJkNGRIZTE...
☑	Content-Type	application/json
	Key	Value

Insert Headers

1) Authorization : Paste Authorization key

2) Content Type : application/json

Getting Start with Postman

- Executing the request of the previous slide (*GET IP_ROBOT/api/v2.0.0/status*), the state of the robot is shown:

Body Cookies Headers (13) Test Results

Status: 200 OK Time: 59ms Size: 1.45 KB Save Response

Pretty Raw Preview Visualize JSON


```
{
    "allowed_methods": null,
    "battery_percentage": 100.0,
    "battery_time_remaining": 136800,
    "distance_to_next_target": 0.0,
    "errors": [],
    "footprint": "[[0.506,-0.32],[0.506,0.32],[-0.454,0.32],[-0.454,-0.32]]",
    "joystick_low_speed_mode_enabled": false,
    "joystick_web_session_id": "",
    "map_id": "8d58d3b9-943d-11ea-9cbc-0800277a1e01",
    "mission_queue_id": null,
    "mission_queue_url": null,
    "mission_text": "Starting...",
    "mode_id": 7,
    "mode_key_state": "idle",
    "mode_text": "Mission",
    "moved": 921.01,
    "position": {
        "orientation": 92.25453186035156,
        "x": 8.340841293334961,
        "y": 13.33425235748291
    },
    "robot_model": "MiR100",
    "robot_name": "MiR_Robot_1",
    "safety_system_muted": false,
    "serial_number": "",
    "session_id": "4ee83dba-8081-11ea-93bf-080027b1ea00",
    "state_id": 4,
    "state_text": "Pause",
    "unloaded_map_changes": false,
    "uptime": 8769,
    "user_prompt": null,
    "velocity": {
        "angular": 0.0,
        "linear": 0.0
    }
}
```


```
"battery_percentage": 100.0,
```

```
"position": {
    "orientation": 92.25453186035156,
    "x": 8.340841293334961,
    "y": 13.33425235748291
},
```

```
"robot_model": "MiR100",
```

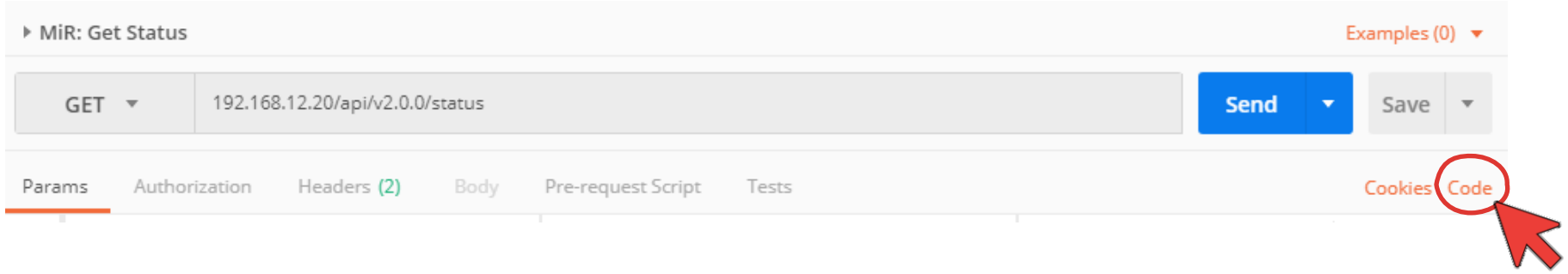
```
"robot_name": "MiR_Robot_1",
```

STATUS OF THE REQUEST

DATA FROM THE ROBOT

MiR RESTful APIs with Postman – Codes

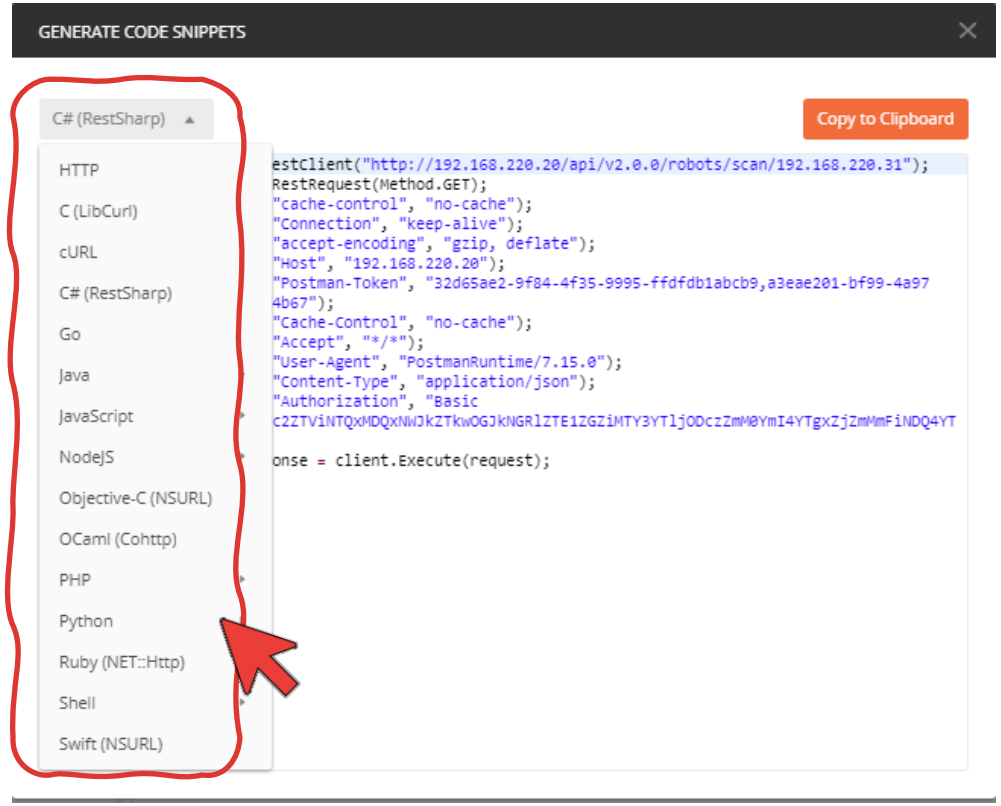
- REST call can be tested and then converted into code directly:



- Select the button labeled **Code** to convert a REST command into another programming language.

MiR RESTful APIs with Postman – Codes

- GET Status command converted into code types can be seen below:



Getting Start with Postman – Codes

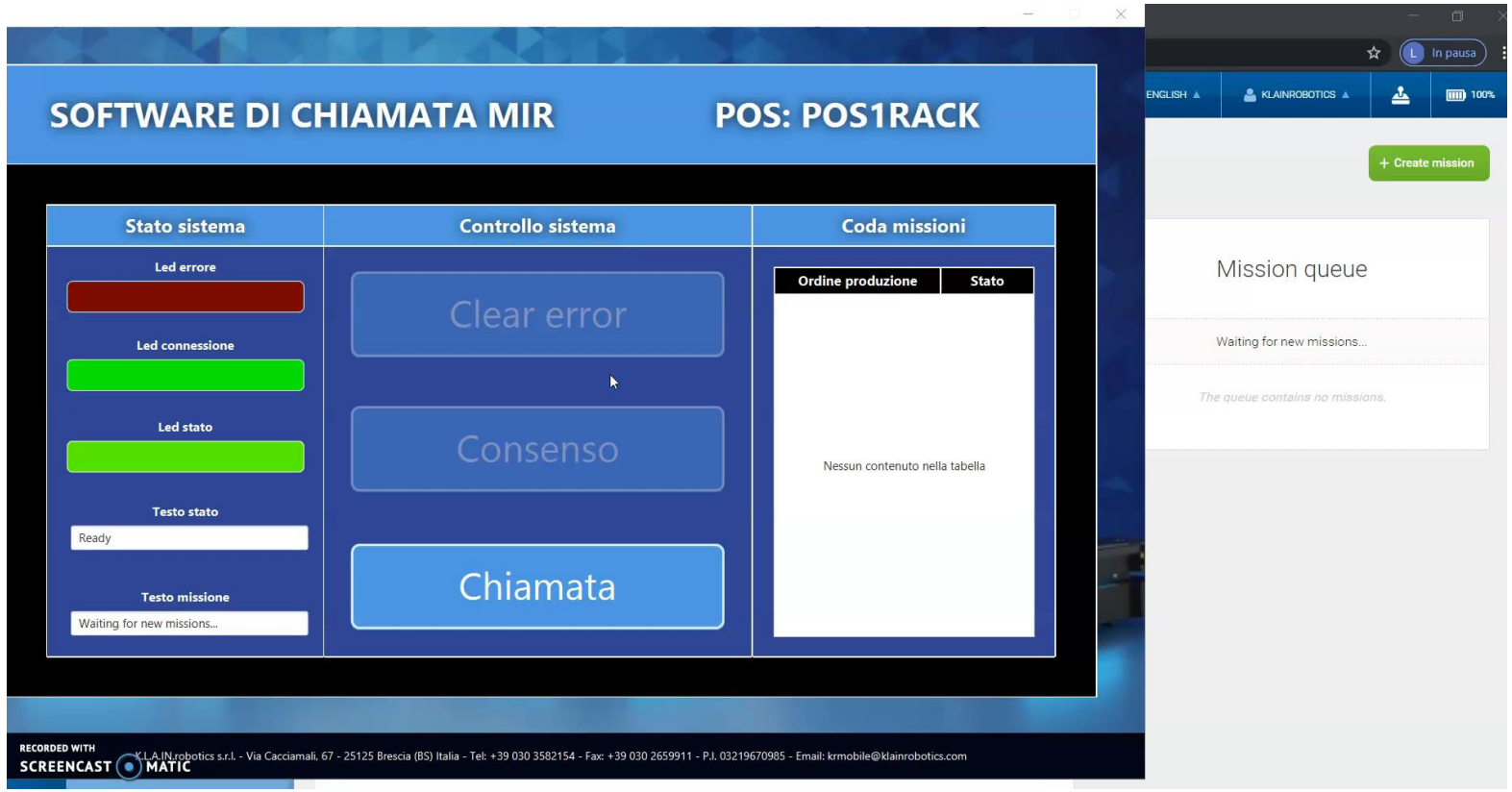
- Copy and Paste onto your scripts

```
var client = new RestClient("http://192.168.220.30/api/v2.0.0/status");
var request = new RestRequest(Method.GET);
request.AddHeader("Content-Type", "application/json");
request.AddHeader("Authorization", "Basic YWRtaW46OGM2OTc2ZTViNTQxMDQxNWJkZTkwOGJkNGRlZTE1ZGZiMTY3YTljODczZmM0YmI4YTgxZjZmMmFiNDQ4YTkxOA==");
request.AddParameter("undefined", "{\r\n  \"mission_id\": \"baf5a091-933f-11e9-bd73-94c691a73638\"\r\n}", ParameterType.RequestBody);
IRestResponse response = client.Execute(request);
```

Real World Example

REST API - Implementation

- Interactions between different points in plant and MiR Robot



Get

MiR

Robot (Check Robot inside the fleet)

[robot IP]/api/v2.0.0/robots

GET http://10.162.160.75/api/v2.0.0/robots

Params Authorization Headers (10) Body Pre-request Script Tests

Query Params

KEY	VALUE
"state_id"	4
Key	Value

Body Cookies Headers (13) Test Results

Pretty Raw Preview JSON


```
[
    {
        "id": 1,
        "url": "/v2.0.0/robots/1"
    },
    {
        "id": 2,
        "url": "/v2.0.0/robots/2"
    },
    {
        "id": 3,
        "url": "/v2.0.0/robots/3"
```


[robot IP]/api/v2.0.0/robots/[id]

GET http://10.162.160.75/api/v2.0.0/robots/1

Pretty Raw Preview JSON


```
            "offset_locked_wheels": 0,
            "width": 0
        },
        "trolley_attached": false
    },
    "joystick_low_speed_mode_enabled": false,
    "joystick_web_session_id": "",
    "map_id": "a58693ee-9283-11e9-ab54-94c691a734b4",
    "mission_queue_id": 0,
    "mission_text": "Waiting for new missions...",
    "mode_id": 7,
    "mode_key_state": "idle",
    "mode_text": "Mission",
    "moved": 16567.152209802865,
    "position": {
        "orientation": 160.1168518066406 2,
        "x": 43.99726486206055,
        "y": 22.125736236572266
    },
    "robot_name": "MiR_R1186",
    "safety_system_muted": false,
    "session_id": "aed92da7-881c-11e9-baa9-94c691a73a68",
    "software_version": "2.7.1",
    "state_id": 3,
    "state_text": "Ready",
    "unloaded_map_changes": false,
    "uptime": 6290,
    "user_prompt": {
        "guid": "",
```

Missions (Check mission GUID)

[robot IP]/api/v2.0.0/missions

GET http://10.162.160.75/api/v2.0.0/missions

Params Authorization Headers (10) Body Pre-request Script Tests

Query Params

KEY	VALUE
"state_id"	4
Key	Value

Body Cookies Headers (13) Test Results

Pretty Raw Preview JSON


```
12 {
13     "guid": "mirconst-guid-0000-0004-actionlist00",
14     "name": "ChargeAtStation",
15     "url": "/v2.0.0/missions/mirconst-guid-0000-0004-actionlist00"
16 },
17 {
18     "guid": "mirconst-guid-0000-0006-actionlist00",
19     "name": "StageAtPosition",
20     "url": "/v2.0.0/missions/mirconst-guid-0000-0006-actionlist00"
21 },
22 {
23     "guid": "mirconst-guid-0000-0005-actionlist00",
24     "name": "Dock",
25     "url": "/v2.0.0/missions/mirconst-guid-0000-0005-actionlist00"
26 },
27 {
28     "guid": "mirconst-guid-0000-0007-actionlist00",
29     "name": "PickupCart",
30     "url": "/v2.0.0/missions/mirconst-guid-0000-0007-actionlist00"
31 },
32 {
33     "guid": "mirconst-guid-0000-0008-actionlist00",
34     "name": "PlaceCart",
35     "url": "/v2.0.0/missions/mirconst-guid-0000-0008-actionlist00"
36 },
37 {
38     "guid": "mirconst-guid-0000-0009-actionlist00",
```

Missions (Check executed mission)

[robot IP]/api/v2.0.0/mission_scheduler

GET 192.168.220.20/api/v2.0.0/mission_scheduler

Body Cookies Headers (13) Test Results

Pretty Raw Preview JSON


```
        "state": "DONE",
        "url": "/v2.0.0/mission_scheduler/34"
    },
    {
        "id": 35,
        "state": "DONE",
        "url": "/v2.0.0/mission_scheduler/35"
    },
    {
        "id": 36,
        "state": "DONE",
        "url": "/v2.0.0/mission_scheduler/36"
    },
    {
        "id": 37,
        "state": "DONE",
        "url": "/v2.0.0/mission_scheduler/37"
    },
    {
        "id": 38,
        "state": "DONE",
        "url": "/v2.0.0/mission_scheduler/38"
    },
    {
        "id": 39,
        "state": "DONE",
        "url": "/v2.0.0/mission_scheduler/39"
    },
```


[robot IP]/api/v2.0.0/missions_scheduler/[id]

GET 192.168.220.20/api/v2.0.0/mission_scheduler/46

Authorization Headers (2) Body Pre-request Script Tests

TYPE

Inherit auth from parent

The authorization header will be automatically generated when you send the request. Learn more about authorization

Body Cookies Headers (12) Test Results

Pretty Raw Preview JSON


```
{
    "allowed_methods": [
        "PUT",
        "GET",
        "DELETE"
    ],
    "created_by": "/v2.0.0/users/mirconst-guid-0000-0005-users0000000",
    "created_by_id": "mirconst-guid-0000-0005-users0000000",
    "created_by_name": "Administrator",
    "description": "",
    "earliest_start_time": "2019-03-29T08:37:49.759284",
    "finish_time": "1970-01-01T01:00:00",
    "id": 46,
    "importance": 0,
    "mission": "/v2.0.0/missions/07afc0f4-4968-11e9-9be0-94c6911ffb63",
    "mission_id": "07afc0f4-4968-11e9-9be0-94c6911ffb63",
    "order_time": "2019-03-29T08:37:49.759298",
    "parameters": {
        "data": []
    },
    "priority": 1,
    "robot_id": 1,
    "start_time": "2019-03-29T08:37:50.164774",
    "state": "EXECUTING"
}
```

Get Robot’s status

GET [robot IP]/api/v2.0.0/status


```
{
    "allowed_methods": null,
    "battery_percentage": 80.5,
    "battery_time_remaining": 77321,
    "distance_to_next_target": 0,
    "errors": [],
    "footprint": "[[0.506,-0.32],[0.506,0.32],[-0.454,0.32],[-0.774,0.15],[-0.774,-0.15],[-0.454,-0.32]]",
    "hook_status": {
        "angle": 0.21148681640625,
        "available": true,
        "braked": true,
        "cart": {
            "height": 0.9100000262260437,
            "id": 18,
            "length": 0.7900000214576721,
            "offset_locked_wheels": 0.10000000149011612,
            "width": 0.699999988079071
        },
        "cart_attached": false,
        "height": 0.38999998569488525,
        "length": 0.6972096562385559
    },
    "joystick_low_speed_mode_enabled": false,
    "joystick_web_session_id": "",
    "map_id": "5567fad0-4bd5-11e9-ad98-0001298f88e2",
    "mission_queue_id": null,
    "mission_queue_url": null,
    "mission_text": "Waiting for new missions...",
    "mode_id": 7,
    "mode_key_state": "idle",
    "mode_text": "Mission",
    "moved": 8488.754073503691,
    "position": {
        "orientation": 2.036212205886841,
        "x": 3.3140697479248047,
        "y": 9.415504455566406
    },
    "robot_model": "MiR200",
    "robot_name": "MiR-S441",
    "safety_system_muted": false,
    "serial_number": "180200011200441",
    "session_id": "21fa5224-4964-11e9-865c-94c6911ffb63",
    "state_id": 4,
    "state_text": "Pause",
    "unloaded_map_changes": false,
    "uptime": 6199,
    "user_prompt": null,
    "velocity": {
        "angular": 0,
        "linear": 0
    }
```


- Whitelist is another way to search specific topic
- Does not require a body as it is part of the Get command
- Getting a single part of the status for example:
  - **GET [robot IP]/api/v2.0.0/status?whitelist=footprint**
  - **GET [robot IP]/api/v2.0.0/status?whitelist=battery_percentage**

Robot Status (Whitelist)

GET [robot IP]/api/v2.0.0/status?whitelist=

GET http://192.168.12.20/api/v2.0.0/status?whitelist=id

Query Params

	KEY	VALUE
☑	whitelist	id
	Key	Value

ody Cookies Headers (10) Test Results

Pretty Raw Preview JSON


```
{
    "available_fields": [
        "joystick_low_speed_mode_enabled",
        "mission_queue_url",
        "mode_id",
        "safety_system_muted",
        "moved",
        "state_id",
        "mission_queue_id",
        "mission_text",
        "footprint",
        "velocity",
        "battery_percentage",
        "state_text",
        "serial_number",
        "robot_name",
        "map_id",
        "user_prompt",
        "uptime",
        "errors",
        "allowed_methods",
        "robot_model",
        "unloaded_map_changes",
        "distance_to_next_target",
        "hook_status",
        "session_id",
        "joystick_web_session_id",
        "mode_text",
        "battery_time_remaining",
        "url",
        "position",
        "mode_key_state"
    ],
    "error_code": "fields not found",
```

Post

Mission scheduler (Send mission via API)

[robot IP]/api/v2.0.0/mission_scheduler

POST http://192.168.220.20/api/v2.0.0/mission_scheduler

Params Authorization Headers (10) Body Pre-request Scr

none form-data x-www-form-urlencoded raw binar

```
1 {
2   "mission_id": "eeccce04-911e-11e9-8348-94c6911ffb63"
3
4 }
```

Specific robot_id

POST http://192.168.220.20/api/v2.0.0/mission_scheduler

Params Authorization Headers (10) Body Pre-request Script

none form-data x-www-form-urlencoded raw binary

```
1 {
2   "mission_id": "eeccce04-911e-11e9-8348-94c6911ffb63",
3   "robot_id": 6
4
5 }
```

Specific time

POST http://192.168.220.20/api/v2.0.0/mission_scheduler

Params Authorization Headers (10) Body Pre-request Script Te

none form-data x-www-form-urlencoded raw binary Gr

```
1 {
2   "mission_id": "eeccce04-911e-11e9-8348-94c6911ffb63",
3   "earliest_start_time": "2019-06-25T010:50:08.096Z",
4   "robot_id": 6
5
6 }
```

Mission scheduler (Post mission for parameterize mission)

Specific position

POST http://192.168.220.20/api/v2.0.0/mission_scheduler

Params | Authorization | Headers (10) | Body | Pre-request Script | Tests

none | form-data | x-www-form-urlencoded | raw | binary | GraphQL BETA | JSON (application/json)

```
1 {
2   "mission_id": "d7c2f763-8c50-11e9-a84f-94c6911ffb63",
3   "earliest_start_time": "2019-06-25T011:50:08.096Z".
4   "parameters": [{"input_name": "position", "value" : "fcec8acc-7579-11e9-b720-94c6911ffb63"}]
5 }
```

Post Mission

GET [robot IP]/api/v2.0.0/missions


```
},
{
    "guid": "520001de-7550-11e9-bd8c-94c6911ffb63",
    "name": "GoToPosition",
    "url": "/v2.0.0/missions/520001de-7550-11e9-bd8c-94c6911ffb63"
}
```


POST [robot IP]/api/v2.0.0/mission_queue

POST http://192.168.220.30/api/v2.0.0/mission_queue

Params Authorization Headers (10) Body Pre-request Script Tests

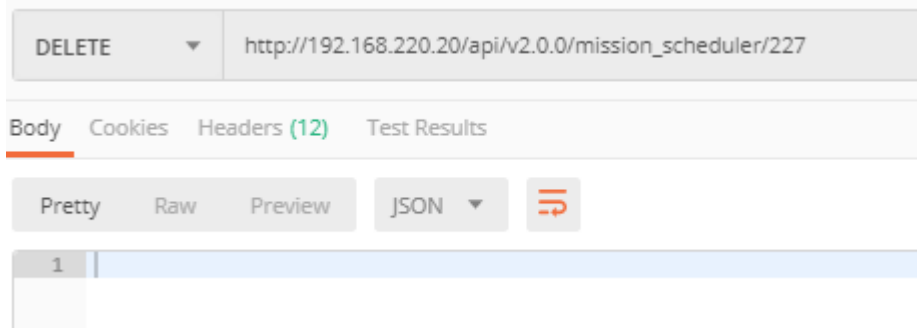
none form-data x-www-form-urlencoded raw binary JSON (application/json)


```
1 {
2     "mission_id": "520001de-7550-11e9-bd8c-94c6911ffb63",
3
4 }
```

Delete

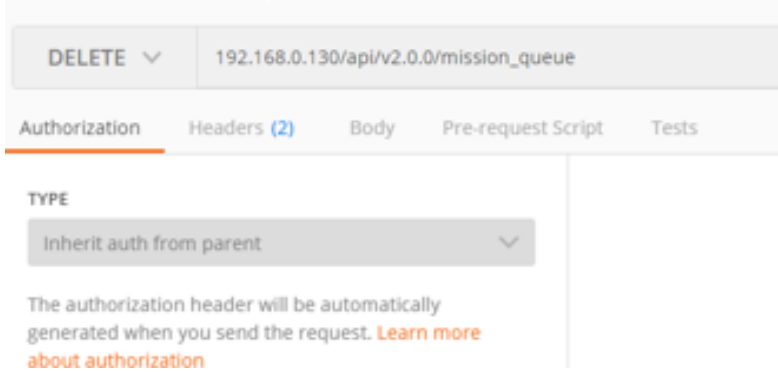
Delete running mission

[robot IP]/api/v2.0.0/mission_scheduler/[id]

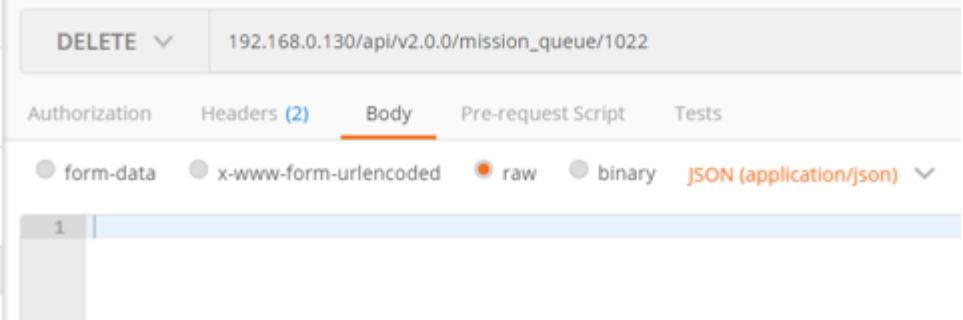


Delete missions (all/ specific ID)

Delete [robot IP]/api/v2.0.0/mission_queue



Delete [robot IP]/api/v2.0.0/mission_queue/[id]



Put

Modify Mission

[robot IP] /api/v2.0.0/mission_scheduler/[id]

PUT http://192.168.220.20/api/v2.0.0/mission_scheduler/228

none form-data x-www-form-urlencoded raw binary

```
1 {
2   "earliest_start_time": "2019-06-25T012:50:08.096Z"
3
4 }
```

Robot Status (How to Play/Pause robot using API)

PUT [robot IP]/api/v2.0.0/status

PUT | http://192.168.220.30/api/v2.0.0/status | Send

Params | Authorization | Headers (2) | Body | Pre-request Script | Tests | Cookies

none | form-data | x-www-form-urlencoded | raw | binary | JSON (application/json)

```
1 {
2     "state_id": 3
3 }
```

State_ID: 3 (continue/play)
4 (Pause)

Change the state of the robot

```
PUT /state
```

Description

Change the state of the robot to pause, ready, etc...

Possible states:

State number	State
0	None
1	Starting
2	Shutting down
3	Ready
4	Pause
5	Executing
6	Aborted
7	Completed
8	Docked
9	Docking
10	Emergency stop
11	Manual control
12	Error

Parameters

Type	Name	Description	Schema
Body	state_id *required*	The id of the new state of the robot	state_id

state_id

Robot Status (How to Clear error)

PUT [robot IP]/api/v2.0.0/status

PUT | 192.168.12.20/api/v2.0.0/status | Send | Save

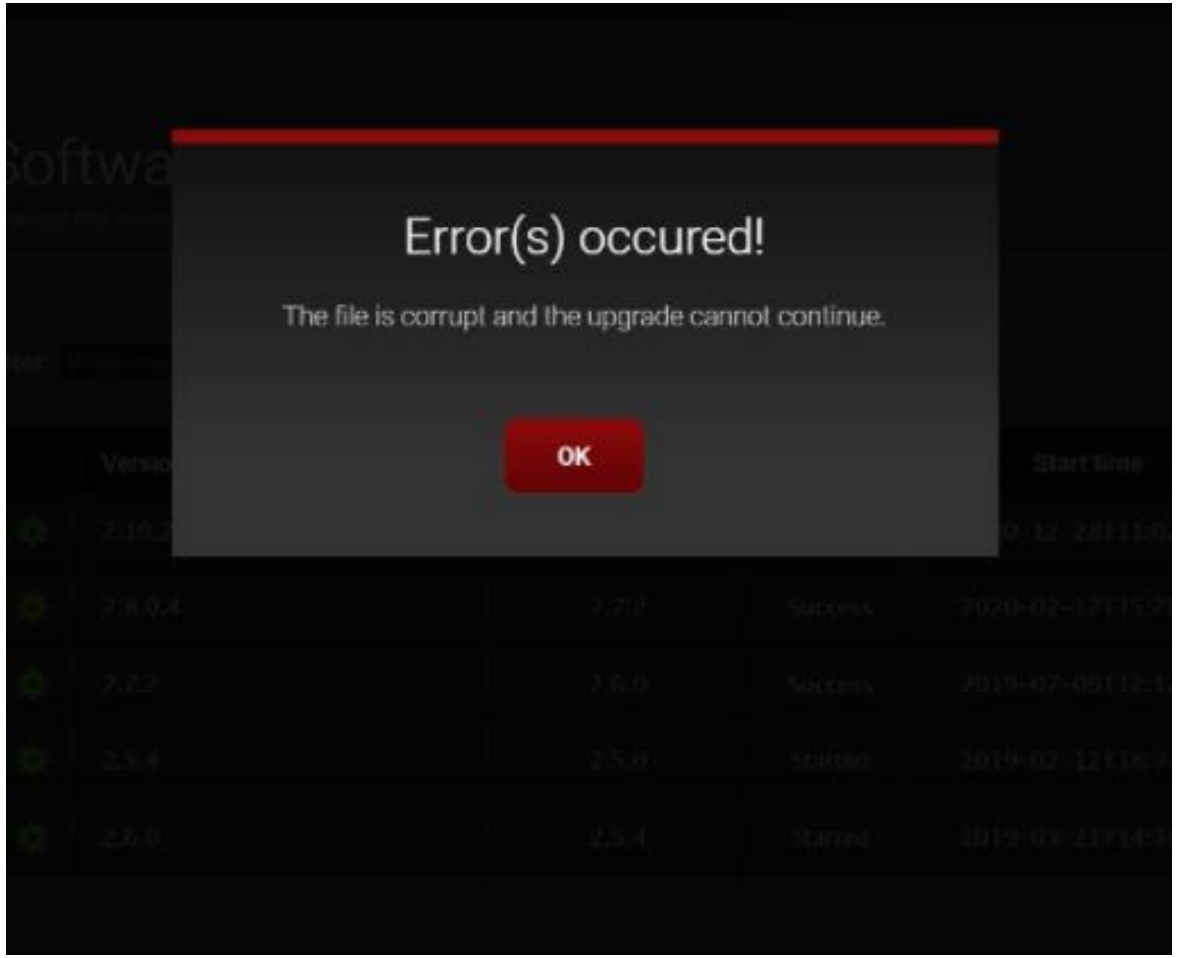
Params | Authorization | Headers (2) | Body | Pre-request Script | Tests | Cookies | Code | Comments (0)

none | form-data | x-www-form-urlencoded | raw | binary | JSON (application/json) | Beautify

```
1 {
2     "clear_error": true
3 }
```

Bonus

Deleting Corrupted Software



Deleting Corrupted Software

- **GET** [robot IP]**/api/v2.0.0/software/upgrades**


```
158   {
159     "url": "/v2.0.0/software/upgrades/af226d26-9860-11eb-83b2-
    0001299c0ea7",
160     "upload_date": "2021-04-08T13:50:59",
161     "guid": "af226d26-9860-11eb-83b2-0001299c0ea7",
162     "version": "MiR_Software-2.10.4.2.mir"
163   }
164 ]
```


- **DELETE** [robot IP]**/api/v2.0.0/software/upgrades/**[id]

DELETE protocol hostname_or_ip /api/v2.0.0/software/upgrades/af226d26-9860-11eb-83b2-0001299c0ea7

JSON Basic Query Header 1 Docs


```
1 {
2
3 }
```